

**IA Test – II**

Program: B.E

Department: CSE

Course Title: Data Structures and Applications

Duration: 1 Hr.

Academic Year: 2024-25 (ODD SEM)

Semester: 3

Max. Marks: 30 marks

Code: 22CS34

Date: 15/11/2024

|               |   |                                                                            |
|---------------|---|----------------------------------------------------------------------------|
| Instructions: | 1 | Missing data, if any, may be suitable assumed.                             |
|               | 2 | Part A is compulsory.<br>Answer any 1 question from Part B and Part C each |

| Q.No                                   | PART A (Compulsory question)                                                                                                                                                                                                                                                                                                                                                                                                                                                            | [L] | [CO] | [PO] | [M] |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------|-----|
| 1a                                     | A person wants to develop a program to illustrate the message passing system using a suitable data structure. His task relates to which course outcome                                                                                                                                                                                                                                                                                                                                  | 1   | 3    | 2,3  | 2   |
| 1b.                                    | <p>a. Define binary search tree. Answer the following with respect to the tree given.</p> <p>a. Root of the tree</p> <p>b. Height of the tree</p> <p>c. Level of the tree</p> <p>d. Balance factor</p> <p>e. Sibling of B</p> <p>f. Degree of D</p> <p>g. Ancestor of G</p> <p>h. Descendants of B</p> <p style="text-align: center;"><b>Tree</b></p>                                                                                                                                   | 2   | 2    | 2,3  | 8   |
| <b>PART B (Answer any 1 questions)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |      |     |
| 2                                      | Develop a program to implement the basic operations of circular queue.                                                                                                                                                                                                                                                                                                                                                                                                                  | 3   | 3    | 2,3  | 10  |
| 3                                      | Develop a C program to traverse an ordered list implemented using a linked list and delete all the nodes whose keys are odd numbers.                                                                                                                                                                                                                                                                                                                                                    | 3   | 3    | 2,3  | 10  |
| <b>PART C (Answer any 1 question)</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |      |      |     |
| 4.                                     | <p>A library management system needs to organize book information using a Binary Search Tree (BST) based on the unique book IDs. Your task is to assist the librarian in arranging the books by inserting the following book IDs into a BST. After the tree is constructed, you will need to develop functions to perform different tree traversals (In-order, Pre-order, and Post-order) and display the books in various orders.</p> <p>58,96,45,76,59,42,31,68,42,25,61,18,20,78</p> | 4   | 4    | 2,3  | 10  |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |     |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|
| 5   | <p>Consider the tree given below as input for questions 5a and 5b.</p> <pre> graph TD     42((42)) --&gt; 28((28))     42 --&gt; 62((62))     28 --&gt; 21((21))     28 --&gt; 31((31))     62 --&gt; 59((59))     62 --&gt; 71((71))     21 --&gt; 17((17))     21 --&gt; 23((23))     31 --&gt; 30((30))     31 --&gt; 35((35))     59 --&gt; 57((57))     59 --&gt; 60((60))     71 --&gt; 67((67))     71 --&gt; 82((82)) </pre> |   |   |     |   |
| 5a. | <p>Analyze the following code snippet and write the output for the given input tree.</p> <pre> struct Node* func(struct Node* node) {     if (node == NULL) {         printf("The tree is empty.\n");         return NULL;     }      struct Node* current = node;     while (current-&gt;left != NULL) {         current = current-&gt;left;     }     return current; } </pre>                                                     | 4 | 3 | 2,3 | 5 |
| 5b. | <p>Analyze the following code snippet and write the output for the given input tree.</p> <pre> struct Node* func (struct Node* node) {     if (node == NULL) {         printf("The tree is empty.\n");         return NULL;     }      struct Node* current = node;     while (current-&gt;right != NULL) {         current = current-&gt;right;     }      return current; } </pre>                                                 | 4 | 3 | 2,3 | 5 |

Pre order - Root    Left    Right  
 In order - Left    Root    Right  
 Post order - Left    Right    Root